# SOWDESHWAR SURVESHA KUMAAR

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## PROFILE

Data scientist with a Master of Data Science from the University of Queensland and two years of earlier data engineering at Vodafone Intelligent Solutions. At Neoware Technology Solutions I work with a global pensions and benefits administrator across its US and UK businesses, on an employee benefits assistant built with LangGraph and AWS AgentCore, a document digitisation and search project where I am the sole technical lead, and a multi-agent analytics assistant on Databricks that includes a time series forecasting feature built with StatsForecast. I also deploy this work with Terraform and Azure DevOps pipelines. My earlier deep learning research with Ericsson was published at IEEE.

## TECHNICAL SKILLS

**GenAI, RAG & Document Intelligence:** RAG, Chunking Strategies, BGE and Titan Embeddings, Qdrant, Amazon OpenSearch, Hybrid Search, Azure Document Intelligence, Amazon Textract, PaddleOCR, Tesseract

**LLMs & Agents:** LangGraph, AWS Bedrock, AgentCore Evaluations and observability, LLM-as-a-Judge, Token and Cost Tracking

**NLP & Text Analytics:** Text Classification, Intent Classification, Word2Vec, Unstructured Document Analysis

**Cloud, Data Platforms & CI/CD:** Azure Databricks, AWS, Azure, Azure DevOps Pipelines, Terraform, Docker, GitHub Actions, MLflow, Git

**Machine Learning & Statistics:** Classification, Regression, Feature Engineering, Cross-Validation, Model Evaluation and Validation, Anomaly Filtering, Concept Drift Detection, Scikit-learn, XGBoost, LightGBM, SHAP, PCA

**Time Series Forecasting:** StatsForecast, ARIMA, ETS, Theta, Rolling-Origin Backtesting, Prediction Intervals, MASE

**Programming & Data:** Python, SQL (Oracle, Spark SQL), Pandas, NumPy, PySpark

**Visualisation:** Power BI, Plotly

## PROFESSIONAL EXPERIENCE

***Neoware Technology Solutions***
**Data Scientist | Jun 2026 - Present | Chennai, India**
*Client: Global pensions, benefits and insurance administrator (BFSI)*

**GenAI Benefits Assistant** | *LangGraph, AWS Bedrock, AgentCore Evaluations, RAG*

- Work on agent orchestration in LangGraph, including intent classification and routing, for an assistant that answers employees' benefits questions from their own plan and enrolment data.
- Own the design of the evaluation and feedback loop, using AWS AgentCore Evaluations with custom LLM-as-a-judge evaluators to score the assistant's responses.
- Researched and compared evaluation tooling (LangSmith, AWS AgentCore Evaluations, Opik, DeepEval) and recommended AgentCore for lower cost, fit with the client AWS agreement and data staying in their VPC. Also researched RAG chunking approaches for plan documents (parent-child, multi-vector, GraphRAG, Contextual Retrieval, Corrective and Adaptive RAG) and prototyped ingestion with Azure Document Intelligence.
- Provisioned the whole assistant as Terraform code through Azure DevOps: user stories, peer review, terraform fmt and validate checks, and a deployment pipeline triggered on merge to main. Docker images go to Amazon ECR and run on AgentCore, with the same code deploying test and production.

**Document Digitisation and Search for a 170 Million Page Archive** | *OCR, Embeddings, Qdrant, AWS*

- Sole technical lead for an NLP and document intelligence project making the archive of scanned pension documents searchable. The design is signed off and the proof of concept is under way.
- Benchmarked open-source and managed OCR engines (Tesseract, PaddleOCR, Surya, Azure Document Intelligence, Amazon Textract, Databricks parsing), 9 embedding models, and vector stores including Qdrant and Amazon OpenSearch.
- Designed tiered, confidence-routed OCR that sends only low-confidence pages to the paid service, after modelling cost and throughput for the full archive.
- Building the proof of concept on about 3,000 real documents (OCR, chunking, Titan embeddings, Qdrant hybrid search), with accuracy measured against the real extracted text, on a PII-compliant AWS design with CI/CD on GitHub Actions.

**Multi-Agent Analytics Assistant** | *LangGraph, Databricks, OpenAI, StatsForecast*

- Own the chart agent in a LangGraph multi-agent assistant that turns business questions into SQL, charts and written summaries. It reads only column metadata and returns validated chart specs, so client rows never reach the LLM.
- Built the forecasting feature end to end and wired it into the orchestrator: planner intent, horizon parsing ("next quarter"), StatsForecast models with 80% prediction intervals, an LLM narrative with a rule-based fallback, and a MASE confidence score shown in the app.

- Benchmarked 10+ forecasting models with rolling-origin backtests on 17 series and real-data holdouts, moving production from AutoTheta to AutoARIMA and replacing a library that took 290 seconds for 24 fits (statsforecast ran 288 fits in 62 seconds).
- Rebuilt the SQL narrative module as a two-stage LLM pipeline with trend-aware sampling, large-result fallbacks, token tracking and handling of negative-signed financial data. Built a package that narrates Power BI exports, switchable between OpenAI and Bedrock.
- **Across all three projects:** prepared cost estimates, infrastructure requirements, Architecture Review Board decks, AI checklists and DPIAs for client sign-off.

***Vodafone Intelligent Solutions (VOIS)***
**Senior Executive, Data and Analytics | Aug 2022 - Jul 2023 | Pune, India**
**Graduate Engineering Trainee | Aug 2021 - Aug 2022**

- Analysed large telecom order datasets with Oracle SQL to find bottlenecks and data quality problems that were breaking downstream reports.
- Redesigned and debugged ETL and integration pipelines (WebMethods) across 3+ source systems, with zero missed SLAs over a 6 month period.
- Added validation checks (schema, referential integrity, null-rate thresholds) so bad data was caught before it reached reports and ML feature tables.
- Automated monitoring of 5+ daily jobs with shell scripts, saving around 10 engineer-hours a week, and supported CI/CD releases through Azure DevOps.

***Ericsson***
**Research Intern, Deep Learning | Dec 2020 - May 2021 | Chennai, India**

- Trained an Inception-ResNet-v2 transfer learning model to classify 15 plant disease classes, reaching 98.77% validation accuracy against 93.1% for a custom CNN and 79.97% for an SVM.
- Built an NLP pipeline that collected farmer remedies from Twitter and ranked them by meaning using Word2Vec and Word Mover's Distance.
- Added seasonal concept drift detection (Bayesian inference with RANSAC) so the system could switch models between seasons without full retraining. Co-authored the published paper.

## PROJECTS

**Noise-Robust ML Pipeline for Large-Scale Yield Prediction** | *Master's Capstone, University of Queensland, 2025*

- Built a distributed pipeline in PySpark and SynapseML that scaled from 100K to 19M+ records with near-linear runtime, about 64 minutes for a full run.
- Benchmarked Random Forest, XGBoost, LightGBM and Bayesian Ridge under 20% to 50% simulated noise, and held $R^2$ at around 0.83 using imputation, PCA denoising and anomaly filtering. Tracked 40+ experiments in MLflow.

## EDUCATION

***University of Queensland (GPA 5.75 / 7)*** — **Brisbane, Australia**
Master of Data Science — **Feb 2024 - Dec 2025**

***Vellore Institute of Technology, Chennai (CGPA 8.24 / 10)*** — **Chennai, India**
B.Tech in Electronics and Communication Engineering — **Jun 2017 - May 2021**

## PUBLICATIONS

- **Advanced Machine Learning Framework for Efficient Plant Disease Prediction.** Co-author. IEEE, ACMLC 2022.
- **Smart Glass with Multi-Functionalities for Assisting Visually Impaired People.** Co-author. Journal of Physics: Conference Series (IOP), ICBSII 2022.

## ACHIEVEMENTS AND VOLUNTEERING

- **Star Employee of the Year,** Vodafone Intelligent Solutions.
- **Winner, Ericsson Design-a-thon,** for a CNN-based pothole detection app, which led to the research internship.
- **Volunteer, Chitlapakkam Rising** (since 2019). Weekly lake cleanups that grew from 10 to 600 volunteers, removed about 60 tonnes of waste and led to Rs. 25 crore (about AUD 4.5 million) in government funding to restore the lake.

## REFERENCES

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